Web Application Vulnerability Scanning with OWASP ZAP

This lab demonstrates **dynamic application security testing (DAST)** using OWASP ZAP against a deliberately vulnerable web application.

Objective

- Perform automated vulnerability scanning.
- Identify common vulnerabilities (XSS, SQL Injection, etc.).
- Interpret alerts and generate a report.

Lab Environment Setup

Step 1: Install OWASP ZAP on Windows

1. Go to the official site
   https://www.zaproxy.org/download/
2. Download the **Windows Installer**
3. Run the installer and follow the setup.

After installation open:

```
OWASP ZAP
```

Step 2: Install Node.js

Juice Shop runs using **Node.js**.

1. Go to
   https://nodejs.org
2. Download **LTS version**
3. Install with default settings.

Step 3: Install OWASP Juice Shop

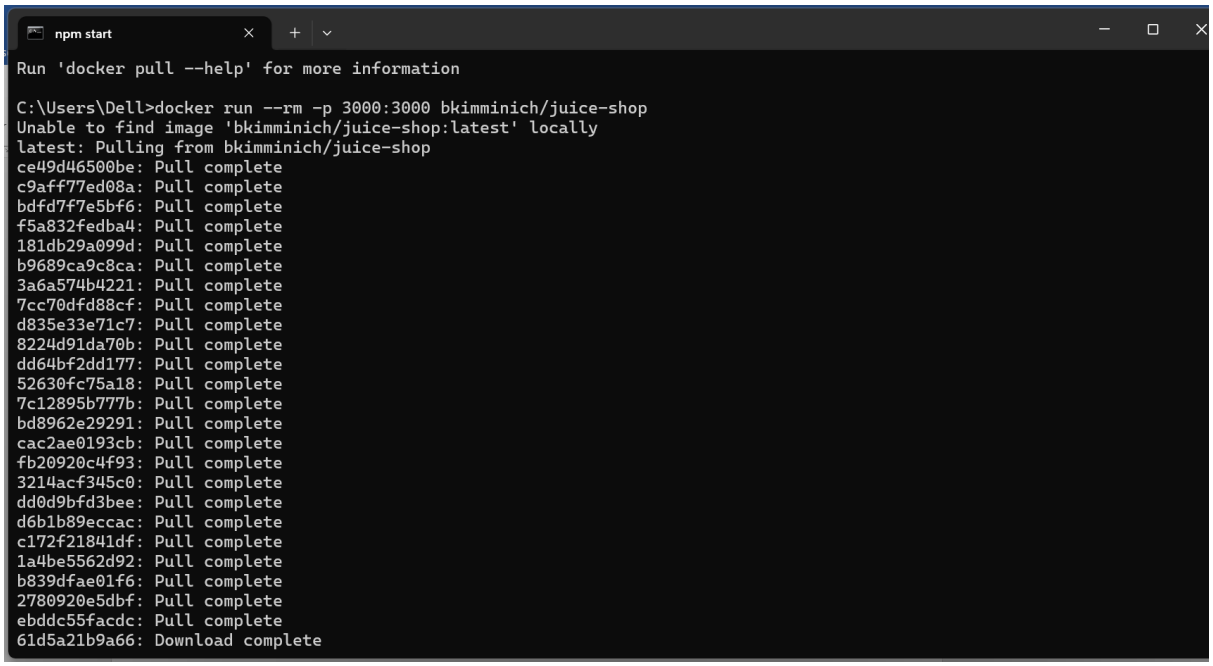
Open **Command Prompt**.

Run:

```
docker run --rm -p 3000:3000 bkimminich/juice-shop
```

OR

```
docker run -d -p 3000:3000 bkimminich/juice-shop
```



Go to the folder:

```
cd juice-shop
```

Install dependencies:

```
npm install
```

Start the application:

```
npm start
```

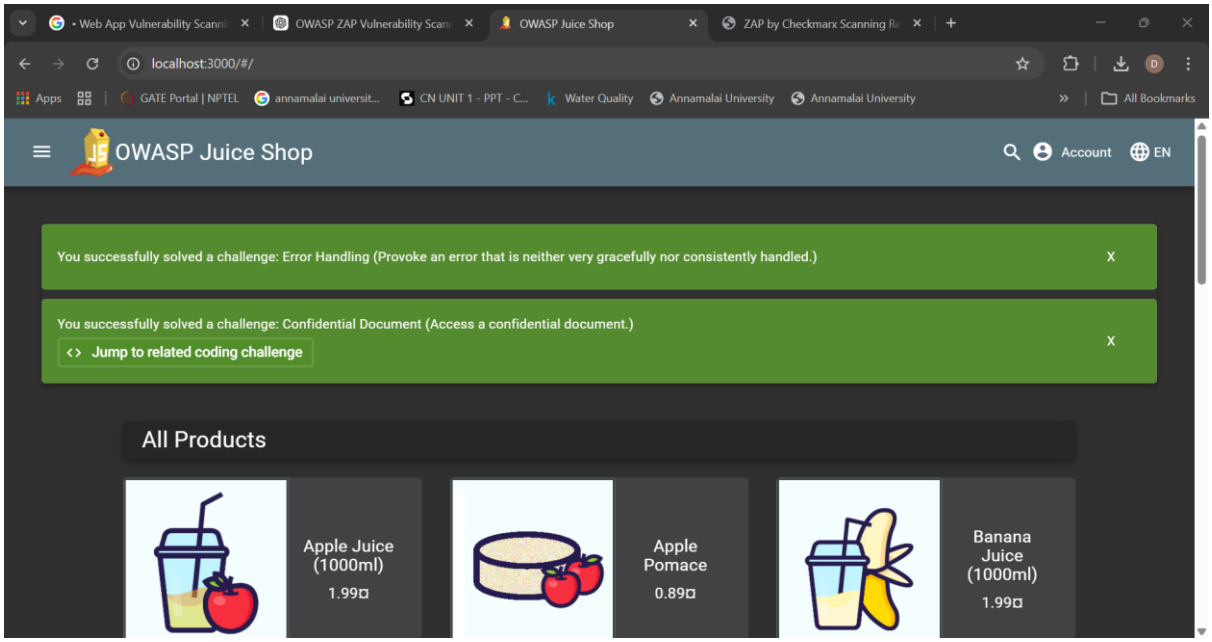
```
npm start
operable program or batch file.

C:\Users\Dell>cd juice-shop

C:\Users\Dell\juice-shop>npm start

> juice-shop@19.2.1 start
> node build/app

info: Detected Node.js version v24.14.0 (OK)
info: Detected OS win32 (OK)
info: Detected CPU x64 (OK)
info: Configuration default validated (OK)
info: Entity models 20 of 20 are initialized (OK)
info: Required file server.js is present (OK)
info: Required file styles.css is present (OK)
info: Required file index.html is present (OK)
info: Required file polyfills.js is present (OK)
info: Required file main.js is present (OK)
info: Required file matching /^hacking-instructor-.+\.js$/ is present (OK)
info: Port 3000 is available (OK)
info: Chatbot training data botDefaultTrainingData.json validated (OK)
info: Domain https://www.alchemy.com/ is reachable (OK)
info: Server listening on port 3000
Error: Only .md and .pdf files are allowed!
    at verify (C:\Users\Dell\juice-shop\build\routes\fileServer.js:59:18)
    at C:\Users\Dell\juice-shop\build\routes\fileServer.js:43:13
    at Layer.handle [as handle_request] (C:\Users\Dell\juice-shop\node_modules\express\lib\router\layer.js:95:5)
    at trim_prefix (C:\Users\Dell\juice-shop\node_modules\express\lib\router\index.js:328:13)
    at C:\Users\Dell\juice-shop\node_modules\express\lib\router\index.js:286:9
```

Step 4: Open Juice Shop

Open browser and go to:

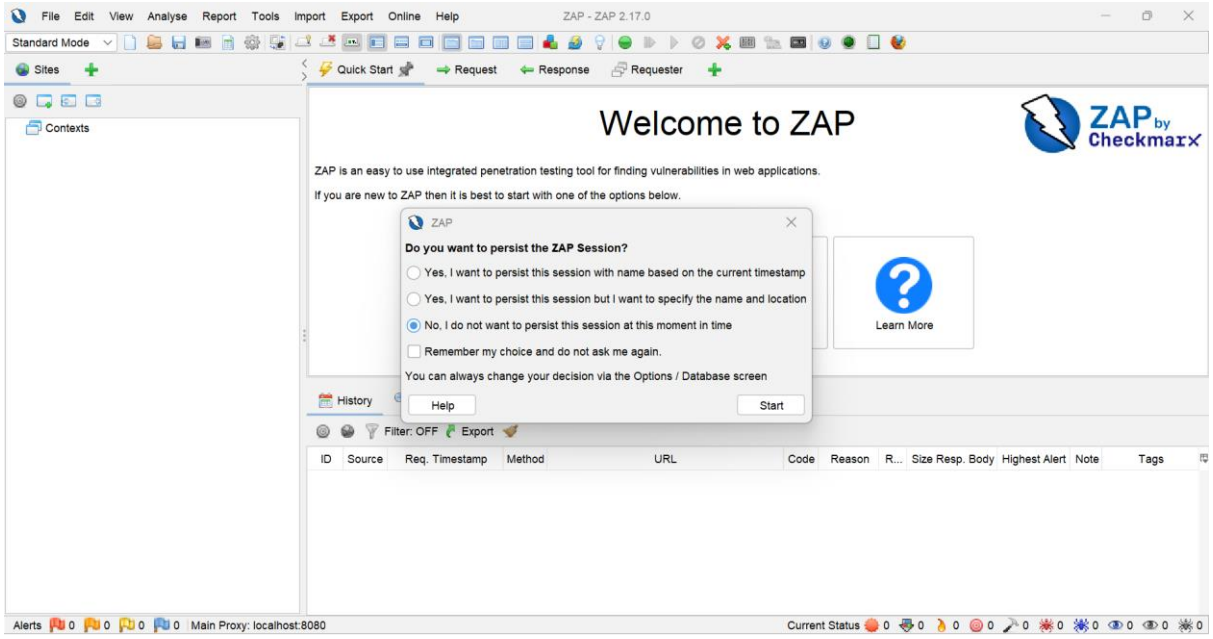
`http://localhost:3000`

You will see the **Juice Shop web application**.



Step 5: Start OWASP ZAP

Open **OWASP ZAP**.



When prompted choose:

`No, I do not want to persist this session`

Click **Start**.

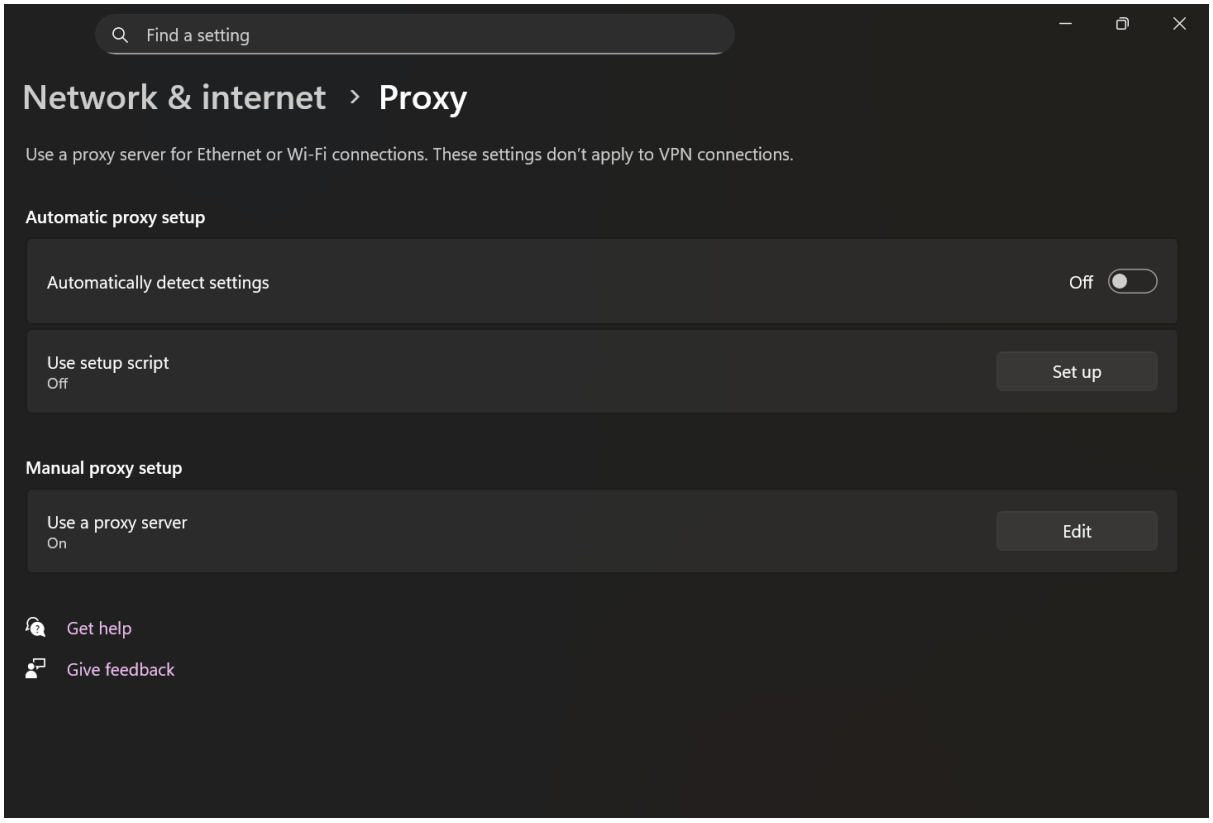
Click on automated scan

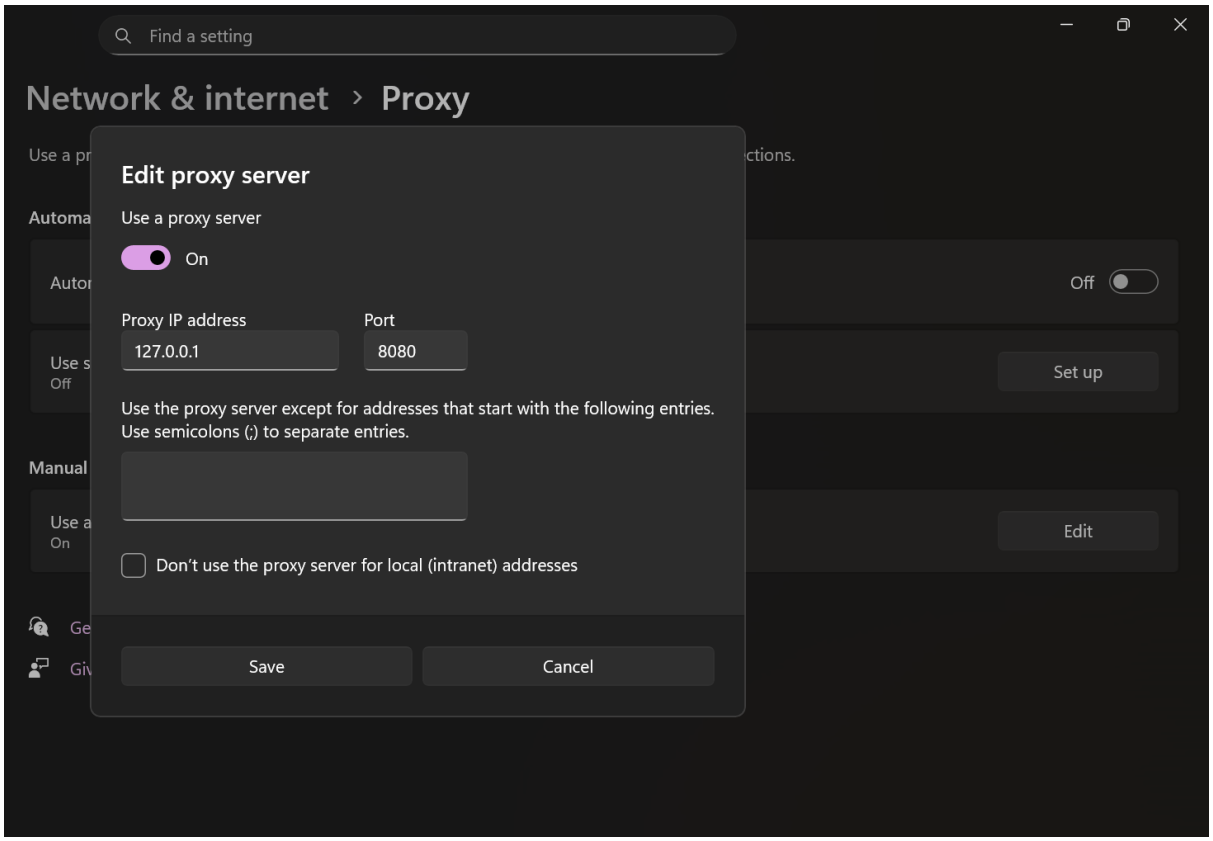
Step 6: Configure Browser Proxy

In browser proxy settings configure:

`HTTP Proxy : 127.0.0.1`

`Port : 8080`





Now browser traffic passes through **ZAP**.

Step 7: Confirm Connection

Open:

`http://localhost:3000`

In ZAP **Sites panel** you will see:

`http://localhost:3000`

Step 8: Perform Spider Scan

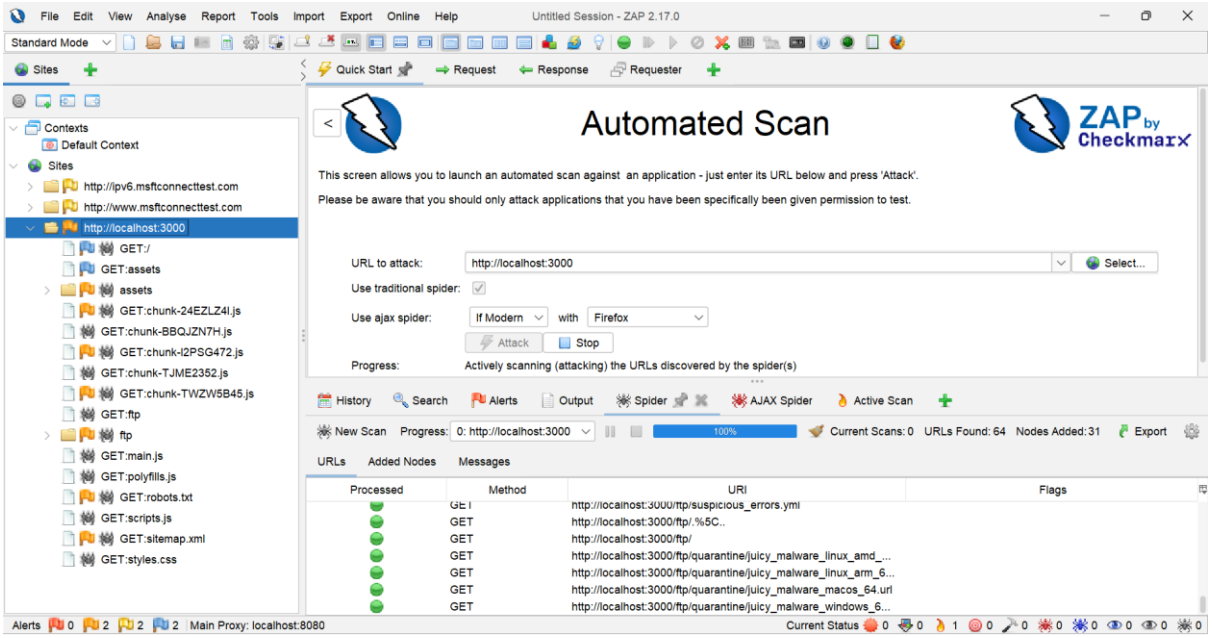
Right-click:

`http://localhost:3000`

Select:

`Attack → Spider`

Click **Start Scan**.



ZAP will discover all pages.

Step 9: Perform Active Scan

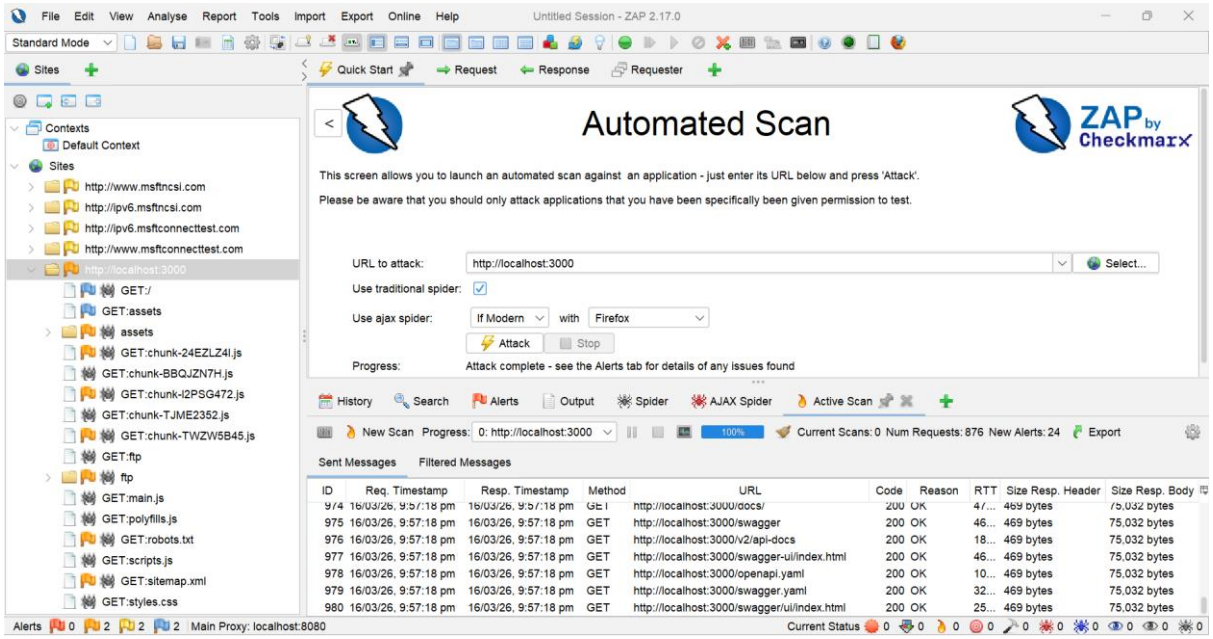
Right-click again:

`http://localhost:3000`

Select:

`Attack → Active Scan`

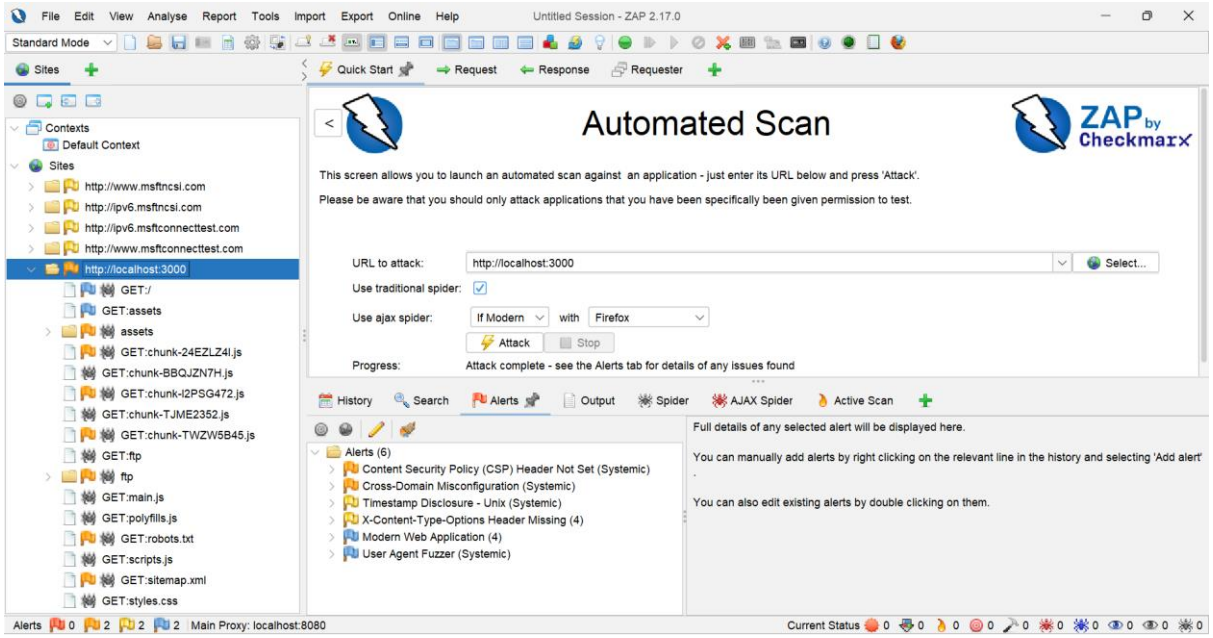
Click **Start Scan**.



ZAP will test vulnerabilities.

Step 10: Analyze Vulnerabilities

Open the **Alerts tab**.



You may see vulnerabilities like:

- Cross Site Scripting
- Missing security headers
- Cookie security issues
- Information disclosure

Analysing Results

Step 11: View Alerts

Go to:

```
Alerts Tab
```

Alerts are categorized by risk:

Risk Level	Meaning
High	Critical vulnerability
Medium	Exploitable issue
Low	Minor issue
Informational	No direct risk

Click any alert to view:

- Description
- Evidence
- Attack payload used
- Suggested remediation

Step 12: Export Scan Report

1. Go to **Report**
2. Choose format:
    - HTML
    - PDF
    - XML
3. Select:
    - Risk levels
    - Confidence level
4. Generate report

Save file.

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Result

The vulnerable web application was scanned using **OWASP ZAP**, and multiple vulnerabilities were identified in **OWASP Juice Shop**.